

Total No. of Questions : 5]

SEAT No. :

PC-2382

[Total No. of Pages : 2

[6354] - 499

B.E. (Computer Engineering)

Deep Learning

(2019 Pattern) (Semester - VIII) (410251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn whenever necessary.*
- 4) *Make suitable assumption whenever necessary.*

Q1) a) How would you describe the architecture of a typical Convolutional Neural Network (CNN)? Explain the main components and their roles. **[6]**

b) What is the purpose of pooling layers in Convolutional Neural Networks? Explain how max pooling and average pooling contribute to down-sampling the input data. **[6]**

c) How do data augmentation and dropout regularization techniques contribute to training CNNs? **[6]**

OR

Q2) a) What role does padding play in Convolutional Neural Networks? How does it impact the size of the output feature maps? **[6]**

b) Explain typical process involved in training a Convolutional Neural Network, including the key steps and optimization techniques used? **[6]**

c) Explain the purpose of LRN in enhancing the performance of CNNs. **[6]**

Q3) a) Explain the concept of unfolding computational graphs in the context of recurrent networks. **[6]**

b) Explain the challenge of long-term dependencies in the context of recurrent neural networks, including the issues of vanishing and exploding gradients. **[6]**

c) What are some common performance metrics used to evaluate the effectiveness of recurrent neural networks? **[5]**

P.T.O.

OR

- Q4)** a) Describe the architecture and usage of Bidirectional RNNs. What advantages do they offer in sequence modeling tasks? [6]
b) What are Leaky Units, and how do they contribute to handling multiple time scales in the context of recurrent neural networks? [6]
c) What are Echo State Networks (ESNs), and how do they address the challenge of long-term dependencies? [5]
- Q5)** a) How do deep generative models differ from discriminative models in terms of their learning and inference mechanisms? [6]
b) How are Deep Belief Networks structured, and how do they leverage the concept of restricted Boltzmann machines in their architecture? [6]
c) What is a Generative Adversarial Network (GAN), and how does it work? [6]

OR

- Q6)** a) Explain the concept of latent variables in deep generative models and their role in capturing complex data distributions. [6]
b) What are the fundamental characteristics of Boltzmann Machines, and how do they model probabilistic dependencies among variables? [6]
c) How can GANs be used in generating realistic images, such as human faces or artistic images? [6]

OR

- Q7)** a) Explain objectives and challenges of deep reinforcement learning in comparison to traditional reinforcement learning methods. [6]
b) Describe the concept of Deep Q-Networks (DQN) and how they combine Q-learning with deep neural networks. [6]
c) Explain how dynamic programming algorithms, such as policy iteration and value iteration, are used in reinforcement learning. [5]

OR

- Q8)** a) What are the key components of a Markov Decision Process, and how do they relate to the decision-making process of an agent? [6]
b) How do Deep Q Recurrent Networks extend the capabilities of Deep Q-Networks in handling sequential decision-making problems? [6]
c) How can reinforcement learning be applied to learn to play Tic-Tac-Toe? [5]

